

PRACTICAL WEEK-3
2520090168
S. Anushka

The image displays two screenshots of the MongoDB Data Explorer interface, showing the 'book_metadata' collection in the 'bookflow' database. The interface includes a sidebar with 'CLUSTERS (1)' and a main panel for document viewing.

Top Screenshot: Shows a single document with the following fields:

```
{
  "_id": ObjectId('6a91432f6d442b6cd5cf3e88'),
  "book_id": 101,
  "title": "The Martian",
  "author": "Andy Weir",
  "published_year": 2021,
  "category": "Science Fiction",
  "format": "Digital PDF",
  "file_size": "2.5 MB",
  "member_id": 101,
  "reviews": Array (3)
}
```

Bottom Screenshot: Shows a document with the following fields:

```
{
  "_id": ObjectId('6a9143ae6d442b6cd5cf3e86'),
  "book_id": 105,
  "title": "Data Structures",
  "author": "Mark Allen",
  "published_year": 2024,
  "category": "Computer Science",
  "format": "Digital PDF",
  "file_size": "5.0 MB",
  "member_id": 105,
  "reviews": Array (2)
}
```

Both screenshots show the 'Documents' tab selected, with a search bar and navigation controls. The bottom screenshot also shows a 'reviews' array field expanded.

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bookflow
 - book_metadata
 - local
 - studentDB
 - students

Cluster0book_metadata

Cluster0 > bookflow > book_metadata

Documents5AggregationsSchemaIndexes1Validation

{ "reviews.rating": { "\$gt": 4 } }

ExplainResetFindOptions

member_id: 101
rating: 5
comments: "Excellent science fiction story"

1: Object (3)
member_id: 102
rating: 4
comments: "Very interesting and informative"

2: Object (3)
member_id: 103
rating: 5
comments: "Excellent space adventure"

Alert System Status: All Good
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My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

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 - students

Cluster0book_metadata

Cluster0 > bookflow > book_metadata

Documents5AggregationsSchemaIndexes1Validation

{ "reviews.rating": { "\$in": [4, 5] } }

ExplainResetFindOptions

0: Object (3)
member_id: 101
rating: 5
comments: "Excellent science fiction story"

1: Object (3)
member_id: 102
rating: 4
comments: "Very interesting and informative"

2: Object (3)
member_id: 103
rating: 5
comments: "Excellent space adventure"

Alert System Status: All Good
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Cluster0

book_metadata

+

Cluster0 > bookflow > book_metadata

Documents 5

Aggregations

Schema

Indexes 1

Validation

{

"format": {

"\$regex": "Digital",

"\$options": "i"

}

}

}

Explain

Reset

Find

Options

25

1-4 of 4

<

>

⋮

{}

⌕

_id: ObjectId('6a91432f6d442b6cd5cf3e80')

book_id: 181

title: "The Martian"

author: "Andy Weir"

published_year: 2021

category: "Science Fiction"

format: "Digital PDF"

file_size: "2.5 MB"

member_id: 181

101

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Cluster0

book_metadata

+

Cluster0 > bookflow > book_metadata

Documents 5

Aggregations

Schema

Indexes 1

Validation

\$match

Explain

Run

Options

Untitled - modified

+ CREATE NEW

PREVIEW

{}

</>

⚙

Stage 1 | \$match

1 {

2 "published_year": {

3 "\$gt": 2020

4 }

5 }

Output preview after \$match stage (Sample of 4 documents)

_id: ObjectId('6a91432f6d442b6cd5cf3e80')

book_id: 181

title: "The Martian"

author: "Andy Weir"

published_year: 2021

category: "Science Fiction"

format: "Digital PDF"

file_size: "2.5 MB"

member_id: 181

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My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

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- local
- studentDB
 - students

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

\$match

\$unwind

View monitoring

Visualize Your Data

Untitled - modified + CREATE NEW

PREVIEW

Stage 2 \$unwind

```
1 /**
2  * path: Path to the array field.
3  * includeArrayIndex: Optional name for
4  * preserveNullAndEmptyArrays: Optional
5  * toggle to unwind null and empty val
6  */
7 {
8   path: "$reviews"
9 }
10
```

Output preview after \$unwind stage (Sample of 10 documents)

```
_id: ObjectId('6a9143ae6d442b6cd5cf3e86')
book_id: 185
title: "Data Structures"
author: "Mark Allen"
published_year: 2024
category: "Computer Science"
format: "Digital PDF"
file_size: "5.0 MB"
member_id: 185
```

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My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bookflow
 - book_metadata
- local
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 - students

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

\$match

\$unwind

\$group

View monitoring

Visualize Your Data

Untitled - modified + CREATE NEW

PREVIEW

Stage 3 \$group

```
1 {
2   "_id": "$title",
3   "average_rating": {
4     "$avg": "$reviews.rating"
5   },
6   "total_reviews": {
7     "$sum": 1
8   }
9 }
```

Output preview after \$group stage (Sample of 4 documents)

```
_id: "Data Structures"
average_rating: 4
total_reviews: 2
```

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My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

bookflow

book_metadata

local

studentDB

students

Cluster0 > bookflow > book_metadata

Documents 5AggregationsSchemaIndexes 2Validation

{

"\$text": {

"\$search": "excellent"

}

}

251-5 of 5

_id: ObjectId('6a91432f6d442b6cd5cf3e80')

book_id: 101

title: "The Martian"

author: "Andy Weir"

published_year: 2021

category: "Science Fiction"

format: "Digital PDF"

file_size: "2.5 MB"

member_id: 101

reviews: Array (3)

Alert System Status: All Good

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ORGANIZATIONPROJECTIXSCAN1/1

Explain Plan

Explain provides key execution metrics that help diagnose slow queries and optimize index usage. [Learn more](#)

Visual TreeRaw Output

```
{
  "parsedTextQuery": {
    "terms": ["excel"],
    "negatedTerms": [],
    "phrases": [],
    "negatedPhrases": []
  },
  "textIndexVersion": 3,
  "inputStage": {
    "stage": "FETCH",
    "inputStage": {
      "stage": "IXSCAN",
      "keyPattern": {
        "_fts": "text",
        "_ftsx": 1
      },
      "indexName": "reviews.comments_text",
      "isMultiKey": true,
      "isUnique": false,
      "isSparse": false,
      "isPartial": false,
      "indexVersion": 2,
      "additionalOptions": {}
    }
  }
}
```

Query Performance Summary

5 documents returned

5 documents examined

0 ms execution time

Is not sorted in memory

5 index keys examined

Query used the following index:

_fts (text)

_ftsx ↑

Close

